

EXPERIMENT 3

Aim:

Create an Employee–Department–Project schema. Insert at least 30 employees across 5 departments and 8 projects. Write SQL queries demonstrating selection, projection, aggregates, GROUP BY, HAVING, CASE expressions, and ORDER BY.

Source Code:

```
CREATE DATABASE company_db;
```

```
USE company_db;
```

```
CREATE TABLE Department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50),  
    location VARCHAR(50)  
);
```

```
CREATE TABLE Project (  
    project_id INT PRIMARY KEY,  
    project_name VARCHAR(100),  
    budget DECIMAL(12,2)  
);
```

```
CREATE TABLE Employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    job_title VARCHAR(50),  
    salary DECIMAL(10,2),  
    hire_date DATE,  
    dept_id INT,  
    manager_id INT,  
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id),  
    FOREIGN KEY (manager_id) REFERENCES Employee(emp_id));
```

```
CREATE TABLE Employee_Project (  
    emp_id INT,  
    project_id INT,
```

```

hours_worked INT,
PRIMARY KEY (emp_id, project_id),
FOREIGN KEY (emp_id) REFERENCES Employee(emp_id),
FOREIGN KEY (project_id) REFERENCES Project(project_id );
INSERT INTO Department VALUES
(1, 'IT', 'Delhi'),
(2, 'HR', 'Gurgaon'),
(3, 'Finance', 'Mumbai'),
(4, 'Marketing', 'Bangalore'),
(5, 'Sales', 'Hyderabad');
INSERT INTO Project VALUES
(101, 'Cloud Migration', 500000),
(102, 'E-Commerce Website', 350000),
(103, 'Mobile Application', 250000),
(104, 'HR Automation', 150000),
(105, 'Financial Analytics', 450000),
(106, 'Marketing Campaign', 200000),
(107, 'CRM Development', 300000),
(108, 'Cyber Security', 600000);
INSERT INTO Employee VALUES
(1, 'Amit', 'IT Manager', 90000, '2018-01-10', 1, NULL),
(2, 'Neha', 'HR Manager', 85000, '2018-03-15', 2, NULL),
(3, 'Rahul', 'Finance Manager', 95000, '2017-06-20', 3, NULL),
(4, 'Priya', 'Marketing Manager', 88000, '2019-02-12', 4, NULL),
(5, 'Arjun', 'Sales Manager', 92000, '2017-09-25', 5, NULL);
INSERT INTO Employee VALUES
(6, 'Ravi', 'Software Engineer', 60000, '2020-01-10', 1, 1),
(7, 'Simran', 'Software Engineer', 65000, '2021-04-15', 1, 1),
(8, 'Karan', 'Database Developer', 70000, '2020-07-20', 1, 1),
(9, 'Pooja', 'Cloud Engineer', 75000, '2022-01-12', 1, 1),
(10, 'Vikas', 'System Engineer', 58000, '2021-09-25', 1, 1),

```

(11, 'Anjali', 'HR Executive', 55000, '2020-02-10', 2, 2),
(12, 'Rohit', 'Recruiter', 50000, '2021-05-15', 2, 2),
(13, 'Kavya', 'HR Executive', 52000, '2022-06-20', 2, 2),
(14, 'Manish', 'Training Officer', 60000, '2020-08-12', 2, 2),
(15, 'Sneha', 'HR Analyst', 58000, '2021-10-25', 2, 2),
(16, 'Deepak', 'Financial Analyst', 70000, '2019-02-10', 3, 3),
(17, 'Nisha', 'Accountant', 62000, '2020-05-15', 3, 3),
(18, 'Varun', 'Financial Analyst', 68000, '2021-06-20', 3, 3),
(19, 'Meena', 'Accountant', 60000, '2022-08-12', 3, 3),
(20, 'Sahil', 'Finance Executive', 57000, '2021-11-25', 3, 3),
(21, 'Tanya', 'Marketing Executive', 55000, '2020-03-10', 4, 4),
(22, 'Mohit', 'SEO Specialist', 60000, '2021-05-15', 4, 4),
(23, 'Isha', 'Content Manager', 65000, '2022-07-20', 4, 4),
(24, 'Akash', 'Marketing Analyst', 58000, '2020-09-12', 4, 4),
(25, 'Riya', 'Social Media Manager', 62000, '2021-12-25', 4, 4),
(26, 'Naveen', 'Sales Executive', 55000, '2020-04-10', 5, 5),
(27, 'Komal', 'Sales Executive', 57000, '2021-06-15', 5, 5),
(28, 'Yash', 'Sales Analyst', 60000, '2022-07-20', 5, 5),
(29, 'Pankaj', 'Business Executive', 65000, '2020-10-12', 5, 5),
(30, 'Divya', 'Sales Executive', 58000, '2022-01-25', 5, 5);

--Assign Employees to Projects

INSERT INTO Employee_Project VALUES

(1,101,100),
(6,101,120),
(7,102,100),
(8,101,90),
(9,108,130),
(10,103,80),
(2,104,100),
(11,104,90),
(12,104,80),

(13,104,70),
(14,104,100),
(15,104,75),
(3,105,100),
(16,105,120),
(17,105,100),
(18,105,90),
(19,105,80),
(20,105,70),
(4,106,100),
(21,106,120),
(22,106,100),
(23,106,90),
(24,106,80),
(25,107,100),
(5,107,100),
(26,107,110),
(27,107,90),
(28,107,80),
(29,107,100),
(30,107,90);

--Selection

Selection means selecting rows according to a condition.

SELECT *

FROM Employee

WHERE salary > 70000;

-- Projection

Projection means selecting specific columns.

SELECT emp_name, job_title, salary

FROM Employee;

--Aggregate Functions

SELECT

```

COUNT(*) AS total_employees,
AVG(salary) AS average_salary,
MAX(salary) AS maximum_salary,
MIN(salary) AS minimum_salary,
SUM(salary) AS total_salary
FROM Employee;
--GROUP BY
SELECT dept_id, COUNT(*) AS employee_count
FROM Employee
GROUP BY dept_id;
--GROUP BY with Average Salary
SELECT dept_id, AVG(salary) AS average_salary
FROM Employee
GROUP BY dept_id;
--HAVING
SELECT dept_id, AVG(salary) AS average_salary
FROM Employee
GROUP BY dept_id
HAVING AVG(salary) > 65000;
-- CASE Expression
SELECT emp_name, salary,
CASE
    WHEN salary >= 80000 THEN 'High Salary'
    WHEN salary >= 60000 THEN 'Medium Salary'
    ELSE 'Low Salary'
END AS salary_category
FROM Employee;
-- ORDER BY
Sort employees according to salary from highest to lowest.
SELECT emp_name, salary
FROM Employee
ORDER BY salary DESC;

```

OUTPUT:

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STDIN

Input for the program (Optional)

Output:

emp_id	emp_name	job_title	salary	hire_date	dept_id	manager_id
1	Amit	IT Manager	90000.00	2018-01-10	1	NULL
2	Neha	HR Manager	85000.00	2018-03-15	2	NULL
3	Rahul	Finance Manager	95000.00	2017-06-20	3	NULL
4	Priya	Marketing Manager	88000.00	2019-02-12	4	NULL
5	Arjun	Sales Manager	92000.00	2017-09-25	5	NULL
9	Pooja	Cloud Engineer	75000.00	2022-01-12	1	1

emp_name	job_title	salary
Amit	IT Manager	90000.00
Neha	HR Manager	85000.00
Rahul	Finance Manager	95000.00
Priya	Marketing Manager	88000.00
Arjun	Sales Manager	92000.00
Ravi	Software Engineer	60000.00
Simran	Software Engineer	65000.00
Karan	Database Developer	70000.00
Pooja	Cloud Engineer	75000.00
Vikas	System Engineer	58000.00
Anjali	HR Executive	55000.00
Rohit	Recruiter	50000.00
Kavya	HR Executive	52000.00
Manish	Training Officer	60000.00
Sneha	HR Analyst	58000.00
Deepak	Financial Analyst	70000.00
Nisha	Accountant	62000.00
Varun	Financial Analyst	68000.00
Meena	Accountant	60000.00
Sahil	Finance Executive	57000.00

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STDIN

Input for the program (Optional)

Output: